

MCS-1: Probability and Statistics (MA6.302)

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Lecture 12

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Machine Learning Lab, IIIT-H

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Total Expectation: Example 2

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- So, we can use the total expectation theorem to get: $\mathbb{E}[M] = \int_{-\infty}^{\infty} \mathbb{E}[M|G = g]f_G(g)dg = \int_5^{15} 30g \times 0.1dg = [1.5g^2]_5^{15} = 300$

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- For two functions $f(X)$ and $g(Y)$, we have $\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)]\mathbb{E}[g(Y)]$.

Independent Random Variable: Example1

- You have two random variables X, Y with joint PDF

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 - $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ for all $x, y \in [0; 1]$
 - X and Y are independent.

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- $c \int_{x=0}^1 \int_0^x xy dx dy = \int_{x=0}^1 x^3 / 2 dx = \frac{c}{8} = 1$. So, $c = 8$.

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- X and Y are not independent.**

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$$f_{XY}(x, y) = \begin{cases} ce^{-2x}e^{-3y}, & x, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

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 - $\mathbb{E}[X] = 1/2$ and $\mathbb{E}[Y] = 1/3$ and via independence $\mathbb{E}[XY] = 1/6$

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 - f_X and f_Y are valid pdfs if $c_1 = 2$ and $c_2 = \frac{1}{\sqrt{2\pi}}$.
- Compute $\mathbb{E}[XY]$
 - Since X and Y are independent, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] = \frac{1}{2} \times 0 = 0$

Independent Random Variables: Example 4

- You have two random variables X and Y with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} ce^{-2x}e^{-\frac{y^2}{2}}, & x \geq 0; y \in \mathbb{R} \\ 0, & \text{otherwise} \end{cases}$$

- Are X and Y independent?

- Joint PDF looks like product of two functions e^{-2x} and $e^{-\frac{y^2}{2}}$.
- Thus, X and Y are independent.

- What is the value of c ?

- Let $f_X(x) = c_1 e^{-2x}$ and $f_Y(y) = c_2 e^{-\frac{y^2}{2}}$.
- f_X and f_Y are valid pdfs if $c_1 = 2$ and $c_2 = \frac{1}{\sqrt{2\pi}}$.
- Thus, $c = c_1 c_2 = \frac{2}{\sqrt{2\pi}}$.

- Compute $\mathbb{E}[XY]$

- Since X and Y are independent, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] = \frac{1}{2} \times 0 = 0$

- The covariance of two random variables X and Y is given by

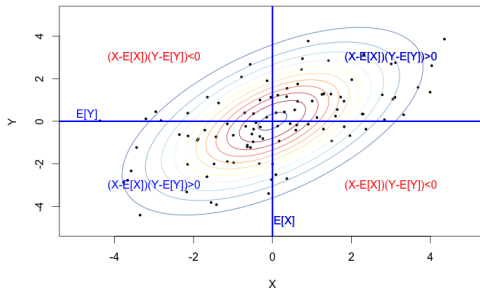
$$\text{cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

- We can simplify this a little

$$\begin{aligned}\text{cov}(X, Y) &= \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \\ &= \mathbb{E}[XY - X\mathbb{E}[Y] - Y\mathbb{E}[X] + \mathbb{E}[X]\mathbb{E}[Y]] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[X]\mathbb{E}[Y] + \mathbb{E}[X]\mathbb{E}[Y] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]\end{aligned}$$

- It is a measure of how much X and Y change together.
- A positive covariance means that, if $X > \mathbb{E}[X]$, we are likely to have $Y > \mathbb{E}[Y]$
- A negative covariance means that, if $X > \mathbb{E}[X]$, we are likely to have $Y < \mathbb{E}[Y]$.

Covariance



- A positive covariance means that we have most mass in the upper right and lower left quadrants.
- A negative covariance means that we have most mass in the upper left and lower right quadrants.
- A zero covariance means that we have about an equal mass in all the quadrants.

Example: Covariance

- Let $f_{X,Y}(x; y) = \begin{cases} 2, & 0 \leq y \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$
- What is $\text{cov}(X, Y)$?
- We first need to calculate the expectation of X and Y .
- The marginal PDF of X and Y are:

$$f_X(x) = \int_0^1 f_{X,Y}(x, y) dy = \int_0^x 2 dy = 2x \quad 0 \leq x \leq 1$$

$$f_Y(y) = \int_0^1 f_{X,Y}(x, y) dx = \int_y^1 2 dx = 2(1 - y) \quad 0 \leq y \leq 1$$

- So the expectations are:

$$\mathbb{E}[X] = \int_0^1 x \cdot 2x \cdot dx = \frac{2}{3}$$

$$\mathbb{E}[Y] = \int_0^1 2(1 - y) dy = \frac{1}{3}$$

Example: Covariance

- We next need to calculate $\mathbb{E}[XY]$.
- This is just the expectation of a function of two random variables

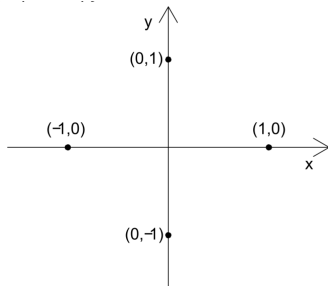
$$\begin{aligned}\mathbb{E}[XY] &= \int_{x=0}^1 \int_{y=0}^x xy 2 dx dy \\ &= \int_0^1 2x \left(\int_0^x y dy \right) dx \\ &= 2 \int_0^1 x \cdot \frac{x^2}{2} dx = \frac{1}{4}\end{aligned}$$

- So, $\text{cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \frac{1}{4} - \frac{1}{3} \times \frac{2}{3} = \frac{1}{36}$

Covariance and Independence

- If two random variables are independent, knowing one tells us nothing about the other!
- In this case, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$
- We know that $\text{cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$. So, if two random variables are independent, their covariance is zero.
- What about the converse? If $\text{cov}(X, Y) = 0$, does that mean that X and Y are independent?

Covariance and Independence



- Let us start at co-ordinates $(0,0)$. I pick a compass direction (N,S,E,W) uniformly at random, and walk 1 unit in that direction.
- Let (X, Y) be my new coordinates. My sample space is $\{(0; 1); (1; 0); (0; -1); (-1; 0)\}$.
- What are $\mathbb{E}[X]$ and $\mathbb{E}[Y]$? 0.
- We also see that $XY = 0$. So, $\text{cov}(X, Y) = 0$.
- But, $X = 1$ implies $Y = 0$. So, they are not independent!
- Independence implies zero correlation... but zero correlation does not imply independence!

Properties of Covariance

- **Symmetric:** $cov(X, Y) = cov(Y, X)$
- $cov(X, X) = Var(X)$
- Let a be a constant and X be a random variable. Then, $cov(X, a) = 0$.
- $cov(X, aY) = a cov(X, Y)$, where a is a scalar.
- $cov(X, Y + b) = cov(X, Y)$, where b is a constant.
- $cov(X_1 + X_2, Y_1 + Y_2) = cov(X_1 + Y_1) + cov(X_1, Y_2) + cov(X_2, Y_1) + cov(X_2, Y_2)$
- $cov(\sum_{i=1}^n a_i X_i, \sum_{j=1}^m b_j Y_j) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j cov(X_i, Y_j)$

- We know that the sign of a covariance indicates whether $X - \mathbb{E}[X]$ and $Y - \mathbb{E}[Y]$ tend to have the same sign.
- The magnitude gives us some indication of the extent to which this is true... but it is hard to interpret.
- The magnitude depends not just how much X and Y co-vary, but also on how much X and Y deviate from their expected values.
- The correlation coefficient $\rho_{X,Y}$ (sometimes referred to as the Pearson's correlation coefficient) is a standardized version of the covariance.

$$\rho_{X,Y} = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X)\text{var}(Y)}}$$

- We always have $-1 \leq \rho_{X,Y} \leq 1$
- $\rho_{X,Y} = 0$ implies zero covariance.
- $|\rho_{X,Y}| = 1$ implies there is a linear relationship between X and Y .

Example: Correlation $|\rho| = 1$

- We throw a biased coin, with probability of heads p , n times.
- Let X be the number of heads, and let Y be the number of tails.
- $Y = n - X$
- $\mathbb{E}[X] = np$, and $\mathbb{E}[Y] = n(1 - p) = n - \mathbb{E}[X]$.
- $\text{var}(X) = np(1 - p) = \text{var}(Y)$.
- $\text{cov}(X, Y) = \text{cov}(X, n - X) = \text{cov}(X, -X) = -\text{cov}(X, X) = -\text{var}(X)$
- The correlation coefficient is therefore

$$\rho_{X,Y} = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X)\text{var}(Y)}} = \frac{-\text{var}(X)}{\sqrt{\text{var}(X)\text{var}(X)}} = -1$$

- Remember $X = n - Y$, so they have a linear relationship.

Example: Correlation $|\rho| = 1$

- Let $Y = aX + b$.
- $\text{cov}(X, aX + b) = \text{cov}(X, aX) = a \cdot \text{var}(X)$
- $\rho_{X, aX+b} = \frac{\text{cov}(X, aX+b)}{\sqrt{\text{var}(X)a^2\text{var}(X)}} = \frac{a \cdot \text{var}(X)}{|a|\text{var}(X)} = \frac{a}{|a|} = \begin{cases} 1, & a > 0 \\ -1, & a < 0 \end{cases}$

More than two continuous random variables

- For multiple random variables we have:

$$P((X, Y, Z) \in B) = \int_{(x,y,z) \in B} f_{X,Y,Z}(x, y, z) dx dy dz$$

- Marginalization: $f_{X,Y}(x, y) = \int_z f_{X,Y,Z}(x, y, z) dz$
- Marginalization: $f_X(x) = \int_y \int_z f_{X,Y,Z}(x, y, z) dy dz$
- Conditional PDF: $f_{X,Y|Z}(x, y|z) = \frac{f_{X,Y,Z}(x, y, z)}{f_Z(z)}$ For $f_Z(z) > 0$
- Conditional PDF: $f_{X|Y,Z}(x|y, z) = \frac{f_{X,Y,Z}(x, y, z)}{f_{Y,Z}(y, z)}$ For $f_{Y,Z}(y, z) > 0$
- Multiplication rule: $f_{X,Y,Z}(x, y, z) = f_{X|Y,Z}(x|y, z) f_{Y|Z}(y|z) f_Z(z)$;
For $f_{Y,Z}(y, z) > 0$
- Independence: $f_{X,Y,Z}(x, y, z) = f_X(x) f_Y(y) f_Z(z)$ For all x, y, z

More than two continuous random variables

- For two random variables X, Y arising out of the same experiment, we define their CDF as:

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u,v) du dv$$

- How do I get $f_{X,Y}(x,y)$ back? $f_{X,Y}(x,y) = \frac{d^2 F_{X,Y}(x,y)}{dx \cdot dy}$
- Let X and Y be jointly uniform on the unit square. $F_{X,Y}(x,y) = xy$ for $0 \leq x, y \leq 1$
- What is $f_{X,Y}(x,y)$?. Differentiate! $\frac{d}{dy} \frac{d}{dx}(xy)$
- This equals 1 for all $0 \leq x, y \leq 1$!

Function of Random Variable

- Let $X \sim \text{Exp}(1)$. Find the PDF of $Y = X^2$.
- How do we do this?
- Start with the CDF. Find $F_Y(y) = P(Y \leq y)$.
- $F_Y(y) = P(X^2 \leq y) = P(X \leq \sqrt{y}) = 1 - \exp(-\sqrt{y})$
- So, $f_Y(y) = \begin{cases} \frac{dF_Y(y)}{dy} = \frac{1}{2\sqrt{y}} \exp(-\sqrt{y}), & y \geq 0 \\ 0, & \text{otherwise} \end{cases}$

Function of Random Variable

- Let $X \sim \text{Uniform}([0, 1])$. Find the PDF of $Y = X^2$.
- Start with the CDF. Find $F_Y(y) = P(Y \leq y)$.
- $F_Y(y) = P(X^2 \leq y) = P(X \leq \sqrt{y}) = \sqrt{y}$
- So, $f_Y(y) = \begin{cases} \frac{dF_Y(y)}{dy} = \frac{1}{2\sqrt{y}}, & y \geq 0 \\ 0, & \text{otherwise} \end{cases}$

Linear Function of Continuous Random Variable: Case 1

- Let X be a random variable with PDF $f_X(x)$, and let $Y = 2X + 3$
- Then the CDF of Y is given by:
$$F_Y(y) = P(2X + 3 \leq y) = P(X \leq \frac{y-3}{2}) = F_X(\frac{y-3}{2})$$
- Differentiating, we get: $f_Y(y) = \frac{dF_Y(y)}{dy} = \frac{d}{dy} F_X(\frac{y-3}{2}) = \frac{1}{2} f_X(\frac{y-3}{2})$

Linear Function of Continuous Random Variable: Case 2

- Let X be a random variable with PDF $f_X(x)$, and let $Y = -2X + 3$
- Then the CDF of Y is given by:
$$F_Y(y) = P(-2X + 3 \leq y) = P(X \geq \frac{-y+3}{2}) = 1 - F_X(\frac{-y+3}{2})$$
- Differentiating, we get:
$$f_Y(y) = \frac{dF_Y(y)}{dy} = -\frac{d}{dy}F_X(\frac{-y+3}{2}) = \frac{1}{2}f_X(\frac{-y+3}{2})$$

Linear Function of Continuous Random Variable: General Case

- Let X be a random variable with PDF $f_X(x)$, and let $Y = aX + b$
- Then the CDF of Y is given by:

$$F_Y(y) = P(aX + b \leq y) = \begin{cases} P(X \leq \frac{y-b}{a}) = F_X(\frac{y-b}{a}), & \text{if } a > 0 \\ P(X \geq \frac{y-b}{a}) = 1 - F_X(\frac{y-b}{a}), & \text{if } a < 0 \end{cases}$$

- Differentiating, we get:

$$f_Y(y) = \frac{dF_Y(y)}{dy} = \begin{cases} \frac{d}{dy} F_X(\frac{y-b}{a}) = \frac{1}{a} f_X(\frac{y-b}{a}), & \text{if } a > 0 \\ -\frac{d}{dy} F_X(\frac{y-b}{a}) = -\frac{1}{a} f_X(\frac{y-b}{a}), & \text{if } a < 0 \end{cases}$$

- $f_Y(y) = \frac{1}{|a|} f_X(\frac{y-b}{a})$

Linear Function of a Normal Random Variable

- Let $X \sim \mathcal{N}(0, 1)$ and $Y = aX + b$
- Then the PDF of Y is given by:

$$\begin{aligned} f_Y(y) &= \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right) \\ &= \frac{1}{|a|} \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-b)^2}{2a^2}} \\ &= \frac{1}{\sqrt{2\pi}|a|} e^{-\frac{(y-b)^2}{2a^2}} \end{aligned}$$

- This is just the PDF of a Gaussian random variable with mean b and variance a^2 .
- In fact, if $X \sim \mathcal{N}(\mu, \sigma^2)$, the $Y \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$

Monotonically Increasing

- We have random variable $X \sim f_X$
- We have $Y = e^X$. What is $f_Y(y)$?

- Start with the CDF.

$$F_Y(y) = P(Y \leq y) = P(e^X \leq y) = P(X \leq \ln y) = F_X(\ln y)$$

- $f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X(\ln y) = \frac{1}{y} f_X(\ln y)$

Monotonically Decreasing

- We have random variable $X \sim f_X$
- We have $Y = e^{-X}$. What is $f_Y(y)$?
- Start with the CDF. $F_Y(y) = P(Y \leq y) = P(e^{-X} \leq y) = P(X \geq -\ln y) = 1 - F_X(-\ln y)$
- $f_Y(y) = \frac{d}{dy} F_Y(y) = -\frac{d}{dy} F_X(-\ln y) = \frac{1}{y} f_X(-\ln y)$

Monotonic Functions of Continuous Random Variables

- Any strictly monotonic function can be inverted, that is there exists an inverse function h (also known as g^{-1}) such that $g(x) = y$ if and only if $x = h(y)$ For all $x \in I$
- If $g(x) = ax + b$, then what is the inverse function?
- Well! Just set $ax + b = y$ and solve for x .
- We get $x = (y - b)/a$. So $x = h(y) = (y - b)/a$.
- The inverse function of $g(x) = e^x$ is $x = h(y) = \ln(y)$
- If g is monotonically increasing, so is h . If g is monotonically decreasing, so is h .

Monotonic Increasing functions of Continuous Random Variable

- Let $Y = g(X)$, where g is monotonically increasing function of X , then $F_Y(y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$.
 - e.g. if $Y = g(X) = e^X$, then $X = g^{-1}(Y) = \ln(Y)$ and $F_Y(y) = P(e^X \leq y) = P(X \leq \ln(y)) = F_X(\ln(y))$
- We can differentiate $F_Y(y)$ to get

$$f_Y(y) = \frac{d}{dy} F_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

- Since g^{-1} is monotonically increasing, it's slope, $\frac{d}{dy} g^{-1}(y)$, must be nonnegative.

Monotonic Decreasing functions of Continuous Random Variable

- Let $Y = g(X)$, where g is monotonically decreasing function of X , then $F_Y(y) = P(g(X) \leq y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$.
 - e.g. if $Y = g(X) = e^{-X}$, then $X = g^{-1}(Y) = -\ln(Y)$ and $F_Y(y) = P(e^{-X} \leq y) = P(X \geq -\ln(y)) = 1 - F_X(-\ln(y))$
- We can differentiate $F_Y(y)$ to get

$$f_Y(y) = \frac{d}{dy} F_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

- Since g^{-1} is monotonically decreasing, it's slope, $\frac{d}{dy} g^{-1}(y)$, must be negative.
- For general strictly monotonic function of X , we have

$$f_Y(y) = \left| \frac{d}{dy} g^{-1}(y) \right| f_X(g^{-1}(y))$$

Example: Monotonically Decreasing Function

- Let X be a uniform random variable on the interval $(0; 1]$, and let $Y = g(X) = \frac{1}{X}$.
- So, Y takes values in $[1, \infty)$
- What is it's inverse? $h(y) = \frac{1}{y}$.
- $f_Y(y) = f_X(h(y)) \left| \frac{d}{dy} h(y) \right|$
- We need to know $f_X(h(y))$ and $\frac{d}{dy} h(y)$
- So, for $y \geq 1$, $f_X(h(y)) = f_X(1/y) = 1$.
- $\left| \frac{d}{dy} h(y) \right| = \left| -\frac{1}{y^2} \right| = \frac{1}{y^2}$
- So, $f_Y(y) = f_X(h(y)) \left| \frac{d}{dy} h(y) \right| = \begin{cases} \frac{1}{y^2}, & \text{for } y \geq 1 \\ 0, & \text{for } y < 1 \end{cases}$

Functions of two random variables

- I know two guests will both arrive in the next hour. The arrival times are independent random variables which follow a uniform distribution. What is the PDF of the arrival time of the last guest to arrive?
- $X \sim \text{Uniform}([0; 1])$ and $Y \sim \text{Uniform}([0; 1])$.
- In other words, what is the PDF of $Z = \max(X; Y)$?
- Let's first think about the CDF...

$$\begin{aligned} P(Z \leq z) &= P(\max(X, Y) \leq z) = P(\{X \leq z\} \cap \{Y \leq z\}) \\ &= P(X \leq z) \cdot P(Y \leq z) = \begin{cases} 0, & z < 0 \\ z^2, & 0 \leq z \leq 1 \\ 1, & z > 1 \end{cases} \end{aligned}$$

- Differentiating,

$$f_Z(z) = \begin{cases} 2z, & 0 \leq z \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

- If $Y = g(X)$, in order to get the PDF of Y we first looked at the CDF, $P(Y \leq y) = P(g(X) \leq y)$ and then differentiated with respect to y .
- For functions $Z = g(X; Y)$ of two random variables, the same general idea applies:
- First we look at the CDF, $P(Z \leq z) = P(g(X; Y) \leq z)$
- Then we differentiate with respect to z .
- We looked at a special case: The maximum of two independent r.v.'s.